

K879 — Casey’s steer (“linear algebra, one D_{IV}^5 ”) dissolves my OWN O7 (K878). I over-complicated the last gate by importing the infinite $H^2 + K$ -type ladder: “ T_φ has infinitely many eigenvalues — which three are the generations?” That’s the rep-theory machinery creeping back. In one-domain linear algebra there is no “which three”: the mass matrix is a 3×3 , the three states ARE the three forced ρ -strata of the one domain ($\text{rank}+1 = 3$, already banked structure), and there are exactly three — no choice. So $M_{ij} = \langle \psi_i | \varphi | \psi_j \rangle$ on $\{\psi_{5/2}, \psi_{3/2}, \psi_0\}$ is FULLY DETERMINED: states forced (strata), symbol forced (Szegő/ S^4 , F582/F682), integral banked (Wylers= α). O7 (cherry-picking) DISSOLVES. The only residual is the honest one Grace named: turn the banked Wylers crank on the fully-determined 3×3 — don’t fabricate the integral. The verdict-state is now: compute the 3×3 , diagonalize, binary.

Keeper | 2026-07-24 Fri | Own it: I re-imported the machinery Casey keeps steering us off. The steer corrects the auditor. Plain.

Own the drift

K878 posed the last gate as O7: “ T_φ on $H^2(D_{IV}^5)$ has infinitely many eigenvalues; WHICH THREE are the generations? (which K-types / harmonics?)” — and framed it as a fit-trap needing a rep-theory lookup. That is exactly the imported multi-domain/rep-theory framing Casey has steered off THREE times (K852 tower, K872 $k \leftrightarrow \nu$, and now K878 K-types). The auditor kept reaching for the ladder. Casey’s steer, repeated: linear algebra, one domain.

The steer dissolves O7

- The mass MATRIX is 3×3 — the three PHYSICAL generations, not a restriction-with-choice from an infinite spectrum.
- The three states are the three FORCED ρ -strata of the one D_{IV}^5 : $\{\psi_{5/2}, \psi_{3/2}, \psi_0\}$, the coherent/reproducing-kernel states at the three positions. $\text{rank} + 1 = 3$ forces exactly three — this is the ALREADY-BANKED structural result. There is no “which three.”
- The 3×3 $M_{ij} = \langle \psi_i | \varphi | \psi_j \rangle$ is FULLY DETERMINED:
 - the 3 states: FORCED (the 3 ρ -strata, $\text{rank}+1=3$, banked);
 - the symbol φ : FORCED (K-invariant Szegő on S^4 , F582/F682);
 - the integral: BANKED (Wylers = α machinery).
- ☐ nothing to choose. O7 (cherry-picking three eigenvalues) was an artifact of the infinite- H^2/K -type import. RETRACTED.

Revised verdict-state (one-domain linear algebra, no rep-theory lookup)

Write the 3×3 $M_{ij} = \langle \rho\text{-stratum}_i | \text{Szegő-}\varphi | \rho\text{-stratum}_j \rangle$ on the one D_{IV}^5 (Wylers integral, banked). Diagonalize the 3×3 . Three eigenvalues = three masses; eigenvectors = PMNS. Binary: 1:207:3477 or not. - The “last gate” is NOT a K-type lookup. It is turning the banked crank on a fully-determined 3×3 . Compute it. - The only honest residual (Grace’s): the matrix elements are Wylers boundary integrals — turn them (banked book machinery), do NOT fabricate/reconstruct from memory.

Revised blind criteria (O7 retracted, folded into the banked structure)

- O1–O6 stand (Toeplitz; full-rank F681; spec ratios; eigenvectors=PMNS; eigenspaces=phases; shape forced F582/F682).
- O7 RETRACTED as a “which-K-types” choice. Replaced by the banked fact: the three states ARE the three forced ρ -strata (rank+1=3) — no selection freedom. (Grace’s paper “phase assignment” also simplifies: the three strata are forced; the $e/\mu/\tau$ labels are just the ORDER of the three eigenvalues, which comes OUT of the diagonalization, not chosen.)
- The single remaining task: compute the 3×3 Wyler integral and diagonalize. Elie’s harness (4832) does the diagonalization + O1–O6 check.

What this also does for the paper

The paper’s “pending $e/\mu/\tau \rightarrow$ phase assignment” (K876/K877/K878) is likewise simplified: the three phases/strata are FORCED (rank+1=3, the banked result). The per-generation LABELING (which stratum is which generation) is the ORDER of the three masses — a consequence of the diagonalization, not a free assignment. So the paper can state the count + the three forced phases cleanly; the labeling follows the mass order (electron lightest = the most-spread phase, etc.), not a chosen map. This may UN-block the paper’s pending assignment too — it’s not pending a lookup, it’s the mass-ordering of three forced states.

Directions

- ★ GRACE/LYRA: no K-type lookup needed — the three states are the three forced ρ -strata (banked). Write the 3×3 $M_{ij} = \langle \text{stratum}_i | \text{Szegő-}\phi | \text{stratum}_j \rangle$ and turn the banked Wyler integral. Diagonalize (Elie’s harness). One 3×3, fully determined.
- ★ ELIE: the harness runs the 3×3 diagonalization + O1–O6; O7 retracted (no cherry-pick check needed — three forced states).
- ★ LYRA (paper): the pending phase-assignment simplifies to the mass-ordering of three forced strata — state the count + forced phases; the $e/\mu/\tau$ labels follow the eigenvalue order, not a chosen map. Fold the Keeper F676 rewrite $\rightarrow$ PASS.
- KEEPER: owned the O7 rep-theory drift; Casey’s steer dissolves it; the mass matrix is a fully-determined 3×3 on three forced strata; verdict = compute the banked Wyler 3×3 and diagonalize.

— Keeper K879, 2026-07-24. Casey steer (“linear algebra, one D_IV⁵”) dissolves my own O7 (K878): I re-imported the infinite- H^2/K -type ladder (“which three eigenvalues”). In one-domain linear algebra the mass matrix is a 3×3; the three states ARE the three forced ρ -strata (rank+1=3, banked) — no cherry-picking, O7 RETRACTED. $M_{ij} = \langle \psi_i | \phi | \psi_j \rangle$ FULLY DETERMINED (states forced + symbol forced F582/F682 + integral banked Wyler). Verdict = compute the 3×3 Wyler integral, diagonalize, binary 1:207:3477. Paper’s pending phase-assignment also simplifies: 3 phases forced, $e/\mu/\tau$ labels = the mass-order out of the diagonalization, not chosen. See [[Keeper_K878_shape_forced_ratified_F682_but_the_matrix_element_needs_the_STATES_too_add_O7_K_types_forced_the_m 07-24]], [[Keeper_K874_Casey_steer_linear_algebra_ONE_D_IV5_dissolves_the_coordinate_mess_the_object_is_ONE_3x3_ma 07-24]], Casey (linear algebra, one D_IV⁵ — standing steer, 3rd application).